Check for updates

Water Resources Research

RESEARCH ARTICLE
10.1029/2021WR031058

# Conjunctive Water Management for Agriculture With Groundwater Salinity

**Yiqing Yao[1], Jay R. Lund[1,2], and Thomas Harter[3]**

[1]Center for Watershed Sciences, University of California, Davis, CA, USA, [2]Department of Civil and Environmental Engineering, University of California, Davis, CA, USA, [3]Department of Land, Air and Water Resources, University of California, Davis, CA, USA

**Key Points:**

- A two-stage stochastic quadratic model estimates the economic impact of groundwater salinity on agriculture, especially with perennial crops
- Economically optimized groundwater management fundamentally changes with groundwater salination
- Artificial recharge for aquifers with severe salt accumulation can reduce both loss of agricultural production and groundwater salinity

**Correspondence to:**
Y. Yao,
yqyao@ucdavis.edu

**Citation:**
Yao, Y., Lund, J. R., & Harter, T. (2022). Conjunctive water management for agriculture with groundwater salinity. *Water Resources Research*, *58*, e2021WR031058. https://doi.org/10.1029/2021WR031058

Received 17 AUG 2021
Accepted 13 SEP 2022

© 2022. The Authors.
This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

**Abstract** Salt accumulations in aquifers significantly affect and transform the conjunctive use of surface water and groundwater supporting irrigated agriculture. Salt accumulates in aquifers under many semi-arid irrigated lands where pumping has lowered water levels enough to prevent drainage of saline groundwater from the basin. This paper provides new insights into optimal conjunctive management of groundwater pumping, recharge, surface water, and cropping patterns with groundwater salinity and hydrologic variability in an irrigated semi-arid region, such as California's western San Joaquin Valley, reducing agricultural crop yields and revenues. A two-stage stochastic quadratic model explores this problem to prescribe economically optimal crop mix and conjunctive water operation policies over a 10-year period with probabilistic annual surface water availability, considering groundwater salinity's harm to crop yields. At low groundwater salinity, agricultural conjunctive use usually pumps most groundwater in drier years, supplied by additional recharge in wetter years. In contrast, at higher groundwater salinity, optimal conjunctive use pumps less in drier years while pumping more in wetter years, when more surface water allows more dilution of saltier groundwater. Reduced pumping in drier years substantially reduces a region's ability to support higher-value perennial crops and reduces or eliminates lower-value annual crops in dry years. Artificial recharge with fresh surface water in wetter years can have economic value from slowing groundwater salination which allows more groundwater use in drier years.

## 1. Introduction

Irrigating with saline water can significantly increase soil salinity due to crop uptake of fresh water while leaving most salts in soil solution, increasing the soil's osmotic potential. Higher osmotic potential in turn decreases the movement of water toward crop roots (Ayers & Westcot, 1985), requiring crops to expend more energy to extract useable water instead of maturing, and thus decreasing crop yields and profits (Allen, 2000). Salts accumulate in soils or in shallow groundwater in arid and semi-arid areas with every cycle of pumping-and-recharge over the irrigation season (Medellín–Azuara et al., 2008; Schoups et al., 2005). Higher evapotranspiration rates and lower precipitation leave salts in the soil, requiring more surface water (also containing salts) to leach soil salinity to groundwater or drainage systems.

When most basin groundwater discharge is pumping for overlying irrigation, with little drainage of salt from the basin, salts imported with surface water and leaching in the basin increase groundwater salinity (Pauloo et al., 2021). This transforms conjunctive use of surface water and groundwater for irrigated agriculture as groundwater becomes saline enough to reduce crop yields. Such groundwater salination is common in the southern and western parts of California's San Joaquin Valley. Nearly two million tons of salt accumulate annually in the San Joaquin Valley, where 250,000 acres of irrigated land have been fallowed and 1.5 million acres are potentially salt-impaired (CV-SALTS, 2017), and lost economic value from salinity could reach $1.2–$2.2 billion/year by 2030 (Howitt et al., 2009). Globally, many irrigated, semi-arid, and arid agricultural regions have similar conditions (Singh, 2014).

Conjunctive use allows some use of more saline groundwater for irrigation (Coe, 1990; Rhoades, 1984). Mandare et al. (2008) studied conjunctive water operations for wheat in a saline groundwater area with scarce surface water supply. Malash et al. (2008) evaluated the effect of saline irrigation on tomato yields and reported that efficient use of low-quality water can leave groundwater salinity unaffected. Other conjunctive water use studies of mixed water qualities of waters for crops such as wheat, maize, and cowpea crops were conducted by

19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Prendergast et al. (1994), Oster and Grattan (2002), Ben-Gal et al. (2009), Kan and Rapaport-Rom (2012), Rasouli et al. (2013), Al Khamisi et al. (2013), and Ortega-Reig et al. (2014).

Modeling conjunctive use considering salinity often uses linear programming (LP) (Kaushal et al., 1985; Singh, 2012; Tyagi, 1986, 1988; Tyagi & Narayana, 1981, 1984). These models typically develop optimal uses of land, crop mix, surface water and saline groundwater for irrigation that maximizes annual profit. However, LP models may produce many equally optimal solutions (Gorelick et al., 1984; Sedki & Ouazar, 2011; Singh, 2014) and under-represent diminishing economic returns for the initially most profitable crops (Hazell & Norton, 1986; Howitt, 1995). This necessitates nonlinear programming (Shang & Mao, 2006; Yeh, 1985). Example models blend poor-quality groundwater and good quality surface water (Gupta et al., 1987; Khan, 1982). Lefkoff and Gorelick (1990a) and Schoups et al. (2006) applied hydroeconomic optimization modeling of an irrigated, saline stream-aquifer system, considering economic, agronomic and hydrologic processes with decisions on crop mix, groundwater pumping and surface water applications.

Conjunctive use with poor-quality water can maintain root zone leaching, reduce salt export to prevent groundwater degradation, and control root zone waterlogging. However, these require careful management of agricultural drainage (Jalali, 2007), which is often difficult. The aforementioned research also maximized crop production rather than profits, which under-appreciated higher value but more salt-sensitive perennial crops. Furthermore, salinity is not often considered as a major factor affecting crop yield and profit, as additional surface water for leaching is usually included as a surface water requirement.

This paper expands two-stage stochastic quadratic optimization modeling for conjunctive water use operations by Marques et al. (2010), Zhu et al. (2015), and Yao (2021) that maximize expected total net economic benefit while considering groundwater salinity effects on crop yields and a range of probabilistic hydrologic events or water year types (WYTs). This approach derives a novel and more reasonable optimal crop mix and conjunctive use operation strategy for farmers while meeting specific intermediate-term (10-year planning horizon) groundwater storage targets for salinizing aquifers (Pauloo et al., 2021) and groundwater salinities ranging from low (500 mg/L) to high (6,000 mg/L), without dynamic climate uncertainty. More details on methods and results appear in Yao (2021).

## 2. Method

### 2.1. Model Formulation

Figure 1 shows the model's structure. Here, "stage" refers to the time-frame of different decisions. First stage decisions (Stage 1) are made at the beginning of each planning horizon (here, $T = 10$ years) and remain constant over these years. First stage decisions include perennial crop acreage, $X_{p,t}$, over a decade $t$, $B1_t$, and the perennial crop establishment cost, $\mathrm{INIP}_t$. The second stage (Stage 2) includes decisions which can vary each year within the planning horizon, responding to each year's hydrologic conditions, WYTs $j$ ($j = 1$ (dry) to 5 (wet)) with probabilities of $p_j$ and surface water amounts $\mathrm{sw}_j$. The hydrologic probabilities are constant within each planning horizon. Stationary hydrology (a fixed probability distribution of surface water availability) is used here as inter-annual variability is typically much larger than changes across 10-year periods. For each probabilistic WYT $j$, the second-stage decisions (contingent on annual hydrologic conditions) are annual crop acreage $X_{a,j,t}$, volumes of groundwater pumping $W_{p,j,t}$, and land area for groundwater recharge $X_{r,j,t}$, resulting in different net expected benefits from annual crop production $B2_t$ and costs of conjunctive water use $\mathrm{EXPC}_t$ over decade $t$.

Boundary conditions for this optimization include groundwater management goals, such as planned groundwater storage depletion (initial groundwater storage $\mathrm{GW}_{t-1}$ > final groundwater storage $\mathrm{GW}_t$), no-overdraft ($\mathrm{GW}_{t-1} = \mathrm{GW}_t$), and planned groundwater storage recovery ($\mathrm{GW}_{t-1} < \mathrm{GW}_t$) in decade $t$, and incoming perennial crop acreage from a previous time period, $X_{p,t-1}$. The management objective is to maximize total net expected benefits over the decade $t$, $Z_t$, which includes the expected net benefit of perennial crops $B1_t$ minus their establishment cost, $\mathrm{INIP}_t$, plus the expected net benefit of annual crops $B2_t$, minus the conjunctive water use operating cost, $\mathrm{EXPC}_t$ (Equation 1).

$$\max Z_t = B1_t(X_{p,t}) - \mathrm{INIP}_t(X_{p,t}, X_{p,t-1}) + B2_t(X_{a,j,t}) - \mathrm{EXPC}_t(X_{r,j,t}, W_{p,j,t}, \mathrm{GW}_{t-1}, \mathrm{GW}_t) \quad (1)$$

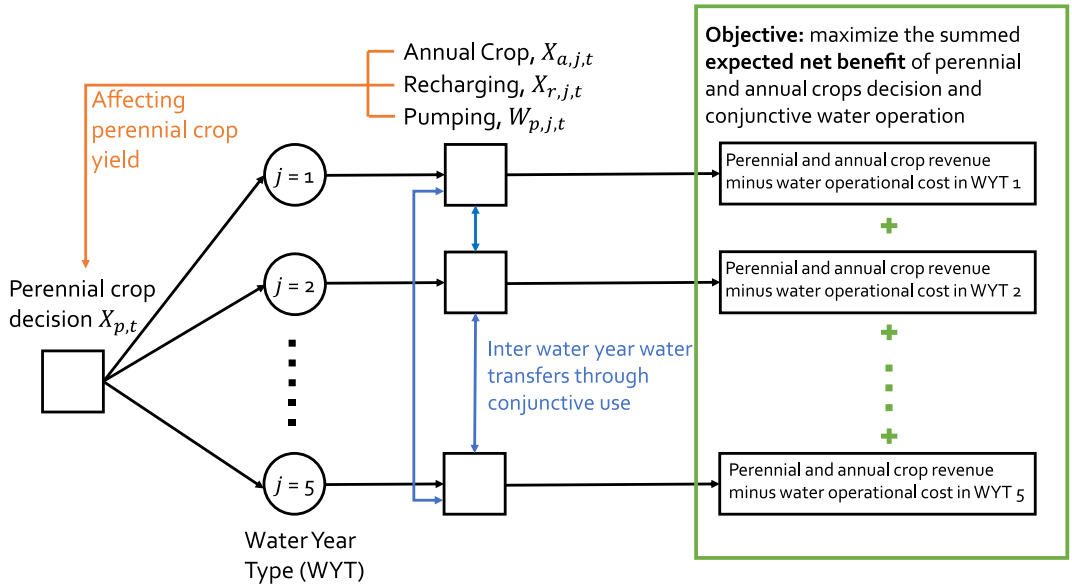



**Figure 1.** Problem decision tree (Marques et al., 2010; Pulido-Velazquez et al., 2016).

If we neglect salinity, $B1_t$ is for the full crop yield rate in all WYTs. However, groundwater salinity affects the salt concentration in irrigation water and perennial crop yield (determining $B1_t$), which now changes with pumping $W_{p,j,t}$ and land use for artificial recharge $X_{r,j,t}$ in each WYT $j$ with its probability, which also is related to annual crop acreage $X_{a,j,t}$ (Equation 3).

Perennial crop yield also is influenced by drought when less surface water is available to dilute more saline groundwater. We assume that if the current year is the driest WYT (WYT 1), no matter which WYT occurs in the next year, the perennial crops are still toxified with lower yields for several years. Detailed derivations of $B1_t$ and $B2_t$ appear in the Appendix A. Derivations of $\mathrm{INIP}_t$ and $\mathrm{EXPC}_t$ are presented in Yao (2021).

The objective function (Equation 1) is subject to resource constraints for land (Equation 2), water (Equation 3), and an expected groundwater mass balance constraint (Equation 4), where $L$ is the total arable area, $cap$ is capacity of land for recharging, $\mathrm{aw}_p$ and $\mathrm{aw}_a$ are annual applied water on perennial and annual crops, respectively, and $\varphi$ is a fixed percentage of water applied to crops that percolates to the underlying aquifer. Detailed explanation of Equations 2–4 appear in Yao (2021).

$$L - X_{p,t} - X_{a,j,t} - X_{r,j,t} \geq 0 \,\forall j \tag{2}$$

$$\mathrm{sw}_j + W_{p,j,t} - \mathrm{cap} \times X_{r,j,t} - \mathrm{aw}_p X_{p,t} - \mathrm{aw}_a X_{a,j,t} \geq 0 \,\forall j \tag{3}$$

$$\mathrm{GW}_t = \mathrm{GW}_{t-1} + T\varphi \mathrm{aw}_p X_{p,t} + T\sum_{j=1}^{5} p_j \varphi \mathrm{aw}_a X_{a,j,t} + T\sum_{j=1}^{5} p_j \mathrm{cap} X_{r,j,t} - T\sum_{j=1}^{5} p_j W_{p,j,t} \tag{4}$$

This paper only considers one hypothetical perennial crop and one hypothetical annual crop, having agronomic parameters mostly based on almonds and alfalfa (CH2MHILL, 2012), to better explore how mixed perennial and annual crop agriculture respond to salinity and climate.

### 2.2. Model Assumptions and Limitations

This analysis has some important underlying assumptions. First, the aquifer is represented as a large well-mixed basin without external drainage. Aquifer recharge occurs only from artificial recharge and deep percolation of excess irrigation carrying all salts from irrigation. During the planning horizon, water level and salinity changes are assumed negligible (i.e., constant pumping cost and salt concentration) regardless of pumping, artificial recharge, and salt leaching during the planning horizon. Also, the aquifer has sufficient initial water storage that the sequence of wet and dry years is irrelevant.



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 1**
*Available Surface Water in Each Water Year Type (WYT)*

| WYT $j$ | Percentile | $sw_j$ (TAF/yr) | $p_j$ |
|---|---|---|---|
| 1 (Dry) | 10th | 248 | 0.2 |
| 2 | 30th | 387 | 0.2 |
| 3 | Median | 526 | 0.2 |
| 4 | 70th | 716 | 0.2 |
| 5 (Wet) | 90th | 1,115 | 0.2 |

We make several simplifying assumptions about salinity: (a) root zone salinity, which affects crop yields, is the same as the irrigation water mix. Although this simplifies soil salinity, soil salinity is strongly correlated to irrigation water salinity; (b) limited surface water availability prevents its use to leach salts in soil to the aquifer, so a leaching factor is omitted. However, deep percolation drains salts from the root zone to the aquifer; (c) water-logging is omitted as this model assumes drainable soils and sufficient depth to groundwater. Even after recharging, the water-table is assumed to not reach the root zone or cause soil salination from capillary rise; (d) salts are perfectly mixed in the aquifer at the end of the planning horizon (see Appendix B) without salt precipitation onto or dissolution from aquifer materials. The salinity relationships in this model could be modified by using regression to predict groundwater salinity changes (Lefkoff & Gorelick, 1990b) or applying models such as HYDRUS to simulate the movement of water and salt transport (Fontanet et al., 2022); (e) the effects of salinity on crop yield is simplified, especially for long-lasting harm to perennial crops. Maas and Grattan (1999) and Grattan (2002) can be applied to model salinity's effects on more crops in California; (f) the same blended mix of groundwater and surface water is applied to both perennial and annual crops. In reality, it is common to have different mixes depending on local conditions, water rights and water sources; and (g) the model can pump more groundwater than required by the crops (detailed in Sections 3.1 and 3.2) without additional disposal cost.

This model also could expand to consider how deficit irrigation interacts with groundwater salinity. This interaction is not considered in the model because (a) perennial crops lose much longer-term yield and profit even using deficit irrigation only in the driest WYT, and (b) though annual crops are more likely to be applied deficit irrigation, their lower profitability leads us to omit this water stress impact. Finally, perennial crop retirement is simplified.

### 2.3. Case Study

The hypothetical agricultural area examined here has an area, $L$, of 500,000 acres, with access to surface water and surface drainage. The underlying aquifer has no significant external groundwater source or drainage. The rechargeable volume of water (20 acre-feet (AF)/acre) corresponds to a specific yield of the unconfined aquifer of 10% with a 200-ft groundwater depth. The 200 vertical feet of aquifer thickness is the aquifer mixing horizon for salinity. Other parameters are summarized in Appendix C, where parameters are derived from CH2MHILL's draft paper (2012) prepared for DWR using Howitt's (1995) method. Surface water availability has a log-normal distribution $\log(sw) \sim N(\mu = 625$ thousand acre-feet (TAF), $\sigma = 400$ TAF), represented as 10th-, 30th-, 50th-, 70th-, and 90th- percentile WYTs. For this base climate (Table 1) each WYT's probability is the same, 20%.

## 3. Results and Discussion

The first results explore how cropping pattern and conjunctive use operation change with different groundwater salinities for an initial groundwater storage ($GW_{t-1} = 10$ MAF) and incoming perennial crop acreage (75,000 acres). Optimized operations are discussed separately for three groundwater storage goals: a balanced "no overdraft," net recovery, and net drawdown. The optimal groundwater pumping strategy changes fundamentally with groundwater salinity and groundwater storage goals.

### 3.1. No Overdraft

The first case balances groundwater storage in over the decade, no overdraft or net recovery, the expected groundwater storage after 10 years equals initial groundwater storage, but storage can vary between years. This implies net groundwater pumping and recharge (including groundwater from deep percolation of irrigation water) reaches an equilibrium by the end of decadal planning horizon. For this groundwater storage goal, increasing salinity brings two major shifts in groundwater pumping (Figure 2): first, total groundwater pumping decreases somewhat from 106 TAF/yr at the lowest groundwater salinity to 90 TAF/yr for the highest groundwater salinity considered (as explained later); and second, groundwater pumping shifts from drier years to wetter years.

At low groundwater salinity, the region pumps groundwater predominantly in drier years to supplement scarce surface water (as in Figure 2, red and yellow columns are highest when groundwater salinity is 500 mg/L). As

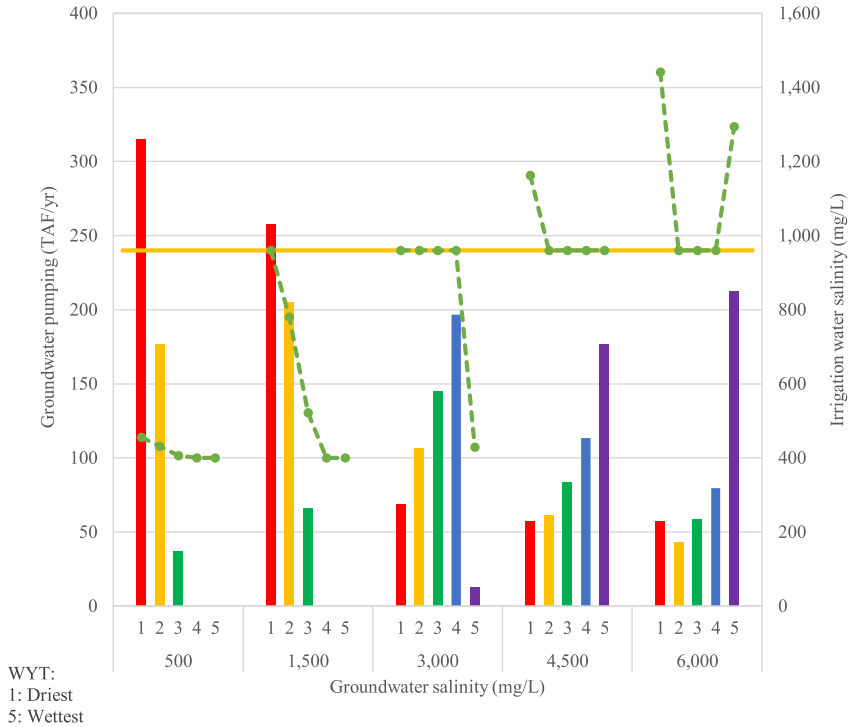



**Figure 2.** Groundwater pumping in different water year types as water year types (WYTs) (columns for each WYT from 1, red, the driest, to 5, purple, the wettest) with different groundwater salinities with "no overdraft" groundwater management. Irrigation water salinity is also shown (green dots) with perennial crops' critical salinity threshold for yield loss (yellow line).

groundwater salinity increases to 1,500 mg/L, higher than the perennial crops' critical salinity (960 mg/L), less groundwater is pumped in the driest WYT (red column shortens), and more pumping occurs in the second driest and intermediate WYTs (yellow and green columns become higher). As groundwater salinities exceed the critical threshold three-fold (3,000 mg/L and above), groundwater pumping further moves to wetter years (red and yellow columns diminish, while the green column first increases then shortens).

The irrigation water salinity exceeding the perennial crops' critical salinity threshold in Figure 2 explains this finding: as groundwater salinity starts to affect crop yields, economically optimized conjunctive use pumps less in drier years, when surface water needed to sufficiently dilute groundwater salinity is lacking. To meet the balanced groundwater storage goal over the planning horizon, more groundwater is pumped in wetter years to where irrigation water salinity hits the perennials' critical level, but does not increase groundwater storage.

The model optimizes groundwater use across WYTs, where perennial crop acreage stays constant (blue columns in Figure 3). When groundwater salinity is low, because more profitable perennial crops dominate, only a small acreage of annual crops in the three drier WYTs (orange columns in Figure 3) (Yao, 2021). Water requirement in the three driest WYTs are the same, so more groundwater pumping occurs in drier years (Figures 2 and 3). The two wetter WYTs have enough surface water to support the entire perennial crop acreage. No groundwater is being pumped and annual crop acreage expands to use all available surface water and increase groundwater recharge from irrigation.

It might be expected that surface water in wet years would be used for artificial recharge, to further increase groundwater availability to grow more perennial crops in the driest WYTs. The model indicates this is not optimal. At low groundwater salinity, the gains from additional perennial crops are less than those of annual crops in wetter WYTs, because of perennial crops' high initial establishment cost and additional cost for artificial recharge in wetter WYTs (Yao, 2021). It is even less economical to recharge surface water when groundwater salinity increases (≥960 mg/L). Even with instantaneous mixing of high-quality surface water recharge with groundwater, pumped groundwater has higher salinity than surface water. Furthermore, irrigation water salinity in drier years already equals or exceeds the critical level, meaning additional pumping in drier years would only reduce perennial crop yield.



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

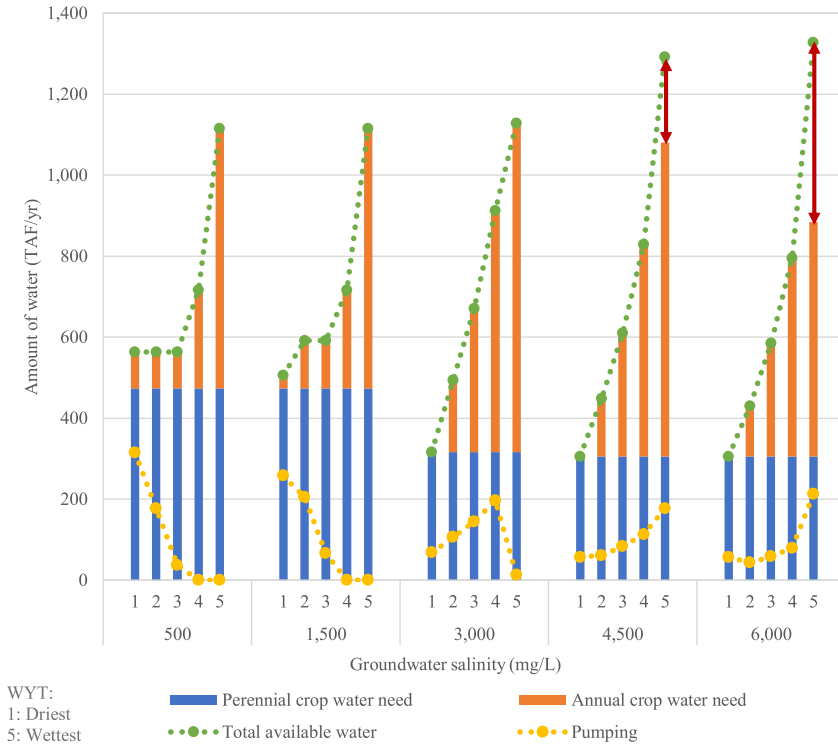



**Figure 3.** Crop water requirements and groundwater pumping decisions under a balanced, "no overdraft" groundwater management goal. Brown arrows in the wettest water year types (WYTs) under high salinities (4,500 mg/L and 6,000 mg/L) indicate the amount of unused irrigation water that must be externally (surface-)drained.

With higher groundwater salinity (≥3,000 mg/L), groundwater pumping greatly decreases in drier years and shifts gradually to wetter years. Irrigating with too much saline groundwater uneconomically decreases perennial crop yields. Perennial crop acreage therefore diminished and is limited by the surface water available in the driest or second driest WYTs to dilute saline groundwater to levels acceptable for perennial crops. Interestingly, there is little difference in perennial crop acreage (and crop water use shown as blue columns in Figure 3) for high salinity conditions (3,000, 4,500, and 6,000 mg/L). This indicates a threshold behavior. Also, no annual crops are grown in the driest years with high salinity conditions (no orange column in WYT 1 in Figure 3), as irrigation water salinity equals or exceeds perennials' salinity threshold (Figure 2).

To further understand this behavior, Figure 4 depicts the model's determination of the perennial crop acreage. When the incoming perennial crop acreage is small relative to available surface water and groundwater salinity is not high, new perennial crops are planted up to an acreage where the marginal increase in profit is less than for annual crops (given high initial perennial crop establishment costs). The new acreage is also limited when irrigation water salinity in the driest WYT exceeds the critical perennial crop threshold. The WYT that limits the perennial crop acreage is called the pivotal WYT. For low salinity (500 mg/L), the pivotal WYT is WYT 1.

However, when incoming perennial crop acreage is large and/or groundwater salinity is enough to harm perennial crop yield, simply fixing the salinity of irrigation water in the driest WYT to the perennial crop's critical salinity threshold causes too much perennial crops retirement and loss in expected profit (unless groundwater salinity precludes perennial crops altogether). Some yield losses are then tolerated. So, WYT 1 is set aside and the second driest WYT (WYT 2) becomes the pivotal WYT.

Once irrigation water salinity in WYT 2 is determined, for a fixed distribution of surface water, we can calculate groundwater pumping and total irrigation water available in WYT 2. Then the perennial crop acreage can be derived: if available irrigation water in WYT 2 exceeds the incoming perennial crop's water requirement, all incoming perennial crop acreage can be kept, and remaining irrigation water goes to annual crops. Otherwise, some incoming perennial crops are retired until perennial crop water needs equal available irrigation water. In this latter case, annual crops are eliminated.



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

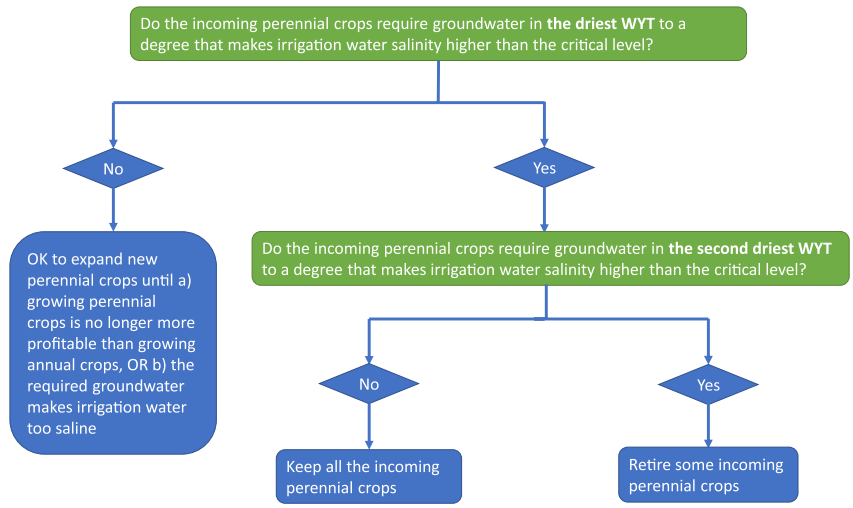



**Figure 4.** Workflow of decision making on perennial crop acreage considering existing groundwater salinity at the beginning of the planning horizon.

The salinity of mixed irrigation water in WYT 1 (driest WYT) already exceeds the perennial crop's critical threshold (Figure 2). It is uneconomical to pump more groundwater for annual crops at the cost of further reducing perennial crop revenue. So groundwater needed in WYT 1 is calculated to support the perennial crop acreage.

With higher salinity, pumping in drier WYTs is limited by surface water availability needed to minimize perennial crop yield impairment. Simultaneously, to maintain groundwater levels, groundwater pumping for irrigation must increase in wetter years, when more surface water exists to dilute saline groundwater.

Pumping in wetter years is also salinity and crop driven: starting from the driest WYT wetter than the pivotal WYT, groundwater should be pumped to grow as many annual crops as possible because of the quadratic characteristic of crop profit, but only to the point when the irrigation water salinity hits the perennial crop's salt tolerance (Figure 2) as irrigation water salinity is the same for both crops, or until all available groundwater (under the no overdraft constraint) is used. Wetter WYTs follow the same steps to determine the pumping amount.

At the highest groundwater salinities (4,500 mg/L and 6,000 mg/L), the wettest WYT shows peculiar important outcomes: some irrigation water (available surface water plus pumped groundwater) is externally discharged to minimize crop losses while achieving 10-year balanced groundwater storage goal. For example, in WYT 5 with groundwater salinity of 4,500 mg/L, irrigation water salinity is kept at the limit for perennial crops (Figure 2) by discharging ("dumping") irrigation water from the system (brown arrow in Figure 3) rather than irrigating annual crops. Using this dumped water to grow annual crops would uneconomically increase recharge via deep percolation. However, in this section, and groundwater needs to maintain storage balance, meaning more pumping is required. More pumping increases irrigation water salinity, and more saline irrigation water reduces perennial crop yield. So fewer annual crops are planted (orange columns in Figure 3), and excess water must be discharged from the basin in the wettest years ("pump and dump," brown arrows in Figure 3). Here, all irrigation water is assumed to have the same groundwater-surface water mixture, so both surface water and groundwater are exported when the "pump and dump" occurs.

When groundwater salinity reaches 6,000 mg/L, the model also makes a trade-off between keeping perennial crops at full yield, but exporting large amounts of irrigation water, and allowing more annual crops but impairing perennial crops in WYT 5 (Figure 3). This dilemma reduces net profit loss by 16% relative to low groundwater salinity conditions (500 mg/L). The loss is underestimated because the decision exploits a loophole in the model formulation, which lacks a penalty for having irrigation water salinity too high in WYTs other than WYT 1 (Figure 2). Also, for cases of larger incoming perennial crop acreage, the negative impact of groundwater salinity is more severe, as larger acreages of incoming perennial crops slow perennial crop retirement, which results in higher irrigation water salinity.

Now let's discuss why Figure 2 shows total expected optimal groundwater pumping decreasing with greater groundwater salinity - when initial groundwater salinity is not excessive (≤3,000 mg/L), groundwater pumping



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

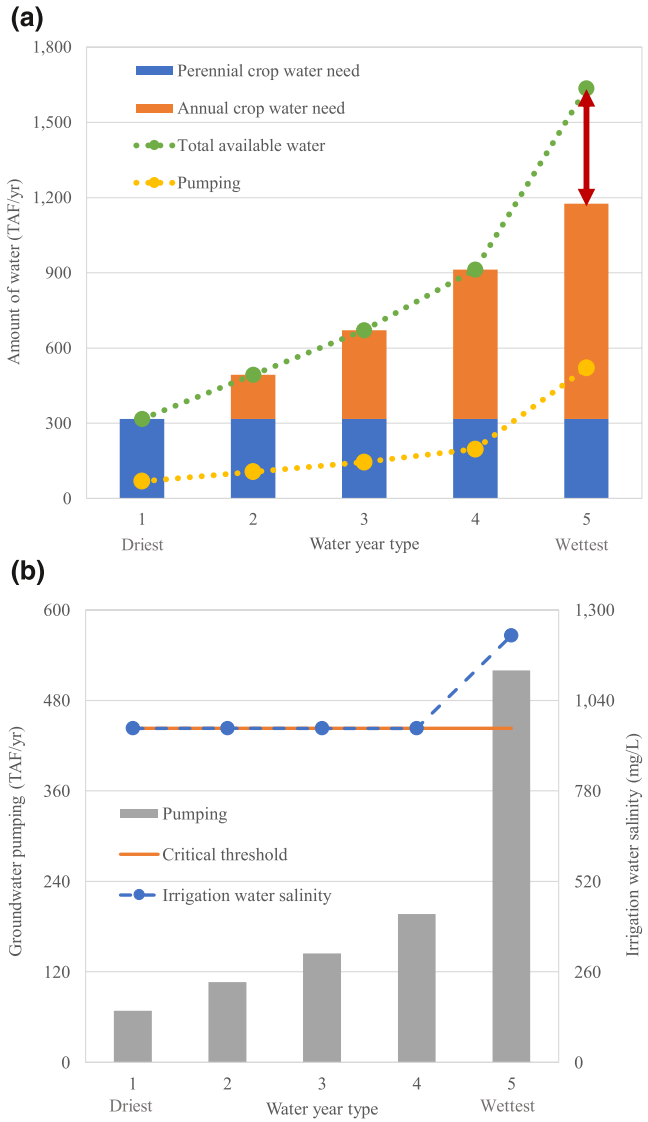



**Figure 5.** Relationship among groundwater pumping, irrigation water requirements, and irrigation water salinity under a net aquifer drawdown management goal (1 MAF over 10 years, equivalent to 0.2 AF/acre/year) with groundwater salinity of 3,000 mg/L. (a) Irrigation water supplied versus crop water needs (b) Irrigation water salinity versus pumping.

is driven purely by surface water availability. Since irrigation efficiency is assumed to be 85%, 15% of irrigation water (combining surface water and pumped groundwater) enters the underlying aquifer, the amount of groundwater pumped is 3/17 of available surface water. However, if initial groundwater salinity is high (≥4,500 mg/L), groundwater pumping is more salinity driven. By restricting annual crop acreage (Figure 3) and deep percolation from irrigation in the wettest WYT, less pumping is needed, and irrigation water salinity is lowered to maximize perennial crop's yield. So, optimal pumping is less when groundwater salinity becomes more severe.

Profit declines with greater groundwater salinity. Though pumping cost is less than for low salinity cases (Figure 2), decreases in crop revenue come first from reduced perennial crop acreage (blue columns in Figure 3) to avoid perennial crop yield loss, which cannot be overcome by increased lower-value annual crop acreage (indeed, in the 6,000-mg/L case, annual crop acreage in WYT 5 is even lower than the 500-mg/L case). Next, as groundwater salinity increases, perennial crops are impaired in the driest WYT (irrigation water salinity in the driest WYT surpasses the critical level in Figure 2), followed by longer-term salinity yield losses.

### 3.2. Net Groundwater Drawdown

Next, we examine optimal conjunctive use and crop planting decisions with planned groundwater depletion—reducing groundwater storage over the planning horizon. A depletion goal is set to 1 MAF over the 10-year planning horizon. At aquifer specific yield of 10%, this implies a basin-wide annual water level decline of 2 ft. The groundwater depletion of 1 MAF becomes available for additional irrigation.

Similar to when the management goal is no groundwater storage change ("no overdraft," Section 3.1), cropping and conjunctive use decisions significantly differ from low salinity to high salinity groundwater conditions. With low salinity conditions, more perennial crops are grown, employing the additional water from storage depletion. When higher groundwater salinities reduce perennial crop yield (e.g., 3,000 mg/L), groundwater is "pumped and dumped" in the wettest WYT (Figure 5a). However, irrigation water salinity in that WYT still exceeds the perennial crop's critical threshold (Figure 5b), so net expected profit for a net drawdown of 1 MAF (1,499 M$) is less than keeping no overdraft (1,581 M$). With high groundwater salinity, overdraft is not economical in this example.

### 3.3. Net Groundwater Storage Recovery

A final set of goals is to gradually increase groundwater storage, perhaps to eventually reconnect groundwater to streams, increase outflow to adjacent groundwater basins, or increase natural salt outflow from the basin, instead of the more expensive "pump and dump" approach (with pumping and treatment costs, although treatment is not considered here).

In contrast to the "no overdraft" management goal, where deep percolation is enough to offset groundwater pumping (which is even too much when groundwater salinity is high), artificial recharge is required under the "net groundwater storage recovery" management goal. With surface water used for recharge, fewer crops are supported, which reduces deep percolation. So less groundwater can be pumped. When groundwater salinity is low, it is economical to recharge surface water by reducing annual crops in wet WYTs to compensate the pumping in drier WYTs for more perennial crop acreage. The cost of recharge and not planting annual crops is less than the income from perennial crops, making this trade-off desirable. Therefore, both pumping and artificial recharge dampen differences in surface water availability for irrigation across years.



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

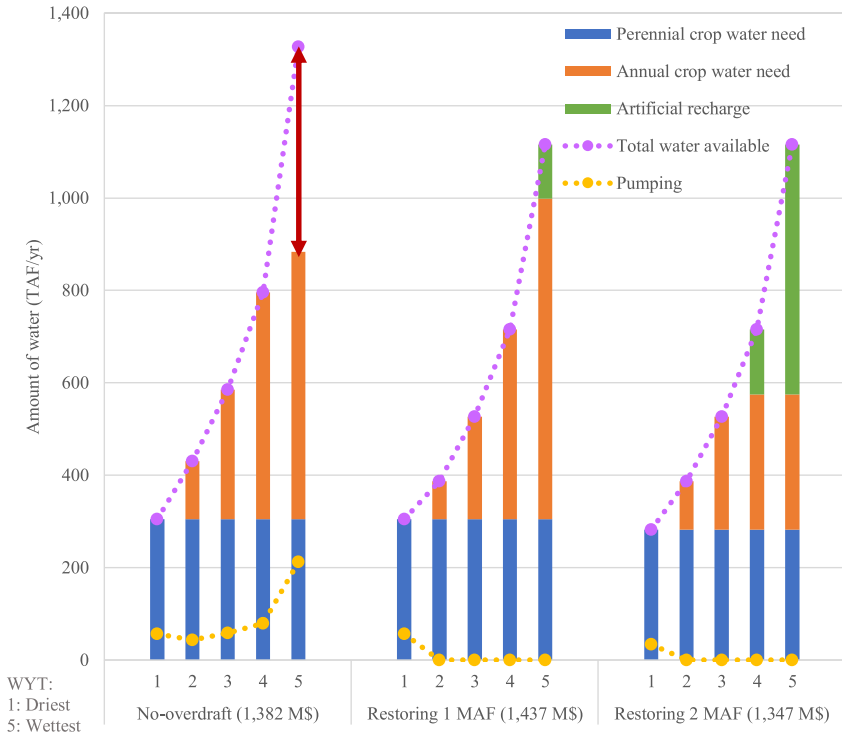



**Figure 6.** With high groundwater salinity (6,000 mg/L), targeting a mild net groundwater recharge within a decade can be more profitable than no overdraft. Artificial recharge dampens surface water variability in wetter water year types (WYTs) and curtails "pump and dump" operations (brown vertical double-arrow in WYT 5 with "no overdraft").

With higher groundwater salinity, pumping is limited in drier years to relieve salinity effects on perennial crops, so pumping no longer dampens year-to-year differences in surface water availability. Artificial recharge occurs in wetter years when crops are only irrigated with fresh surface water. Because of diminishing marginal profit for the annual crop, its acreage should be the same across years with artificial recharge (e.g., restoring 2 MAF in Figure 6), meaning artificial recharge still smooths surface water use for irrigation across wetter years.

For the most saline conditions, groundwater storage recovery of 1 MAF within the planning horizon supports the same acreage of perennial crops as with "no-overdraft" management (blue columns have same height in Figure 6). If waterlogging from rising water levels, and damage to crop roots and soil quality are not an issue, the groundwater storage recovery reduces the need to "pump and dump" saline groundwater (Figure 6), which also possibly requires large treatment and disposal cost. Groundwater recharge also reduces restrictions on planting annual crops in wetter years (the orange column in WYT 5 of net 1-MAF recharge is higher than in the no-overdraft case in Figure 6), as irrigation return flow from the additional wetter year annual crop acreage increases net recharge and does not lead to additional pumping in these WYTs that would affect irrigation water salinity. A higher net recharge goal of 2 MAF slightly decreases perennial crop acreage, but annual crop acreages are significantly reduced in wet years to supply additional artificial recharge. As a result, overall profits under the more aggressive 2 MAF net groundwater storage recovery goal are less than with a mild net groundwater storage recovery goal of 1 MAF over the 10-year planning horizon. The outcome that a net recovery of 1 MAF is most profitable for high groundwater salinity may be different if perennial crops and annual crops are allowed to use irrigation water of different qualities, a condition not considered here.

Unlike the balanced "no-overdraft" management goal, where perennial crop acreage and pumping decisions are driven by irrigation water salinity (Section 3.1, Figure 4), net aquifer recovery management leads to a more complex decision framework: pumping in drier years and perennial crop acreage are not entirely based on irrigation water salinity (still true for a net recharge of 1 MAF), because reduced groundwater availability over the planning horizon can outweigh salinity considerations (e.g., a net recharge of 2 MAF). Pumping in drier years is always less with high groundwater salinities to limit yield loss in perennial crops, which reduces needs for managed aquifer recharge in wetter years (Figure 7), so more annual crops to be planted that contribute return



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

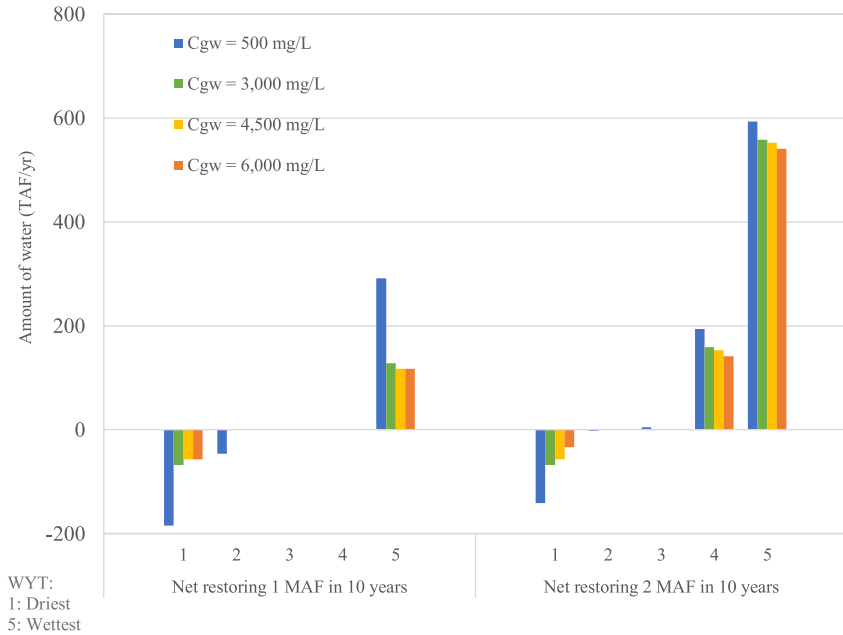



**Figure 7.** Optimal pumping (shown in negative values) and artificial recharge (shown in positive values) for each water year type (WYT) given different net recharge goals and groundwater salinities.

flow to groundwater (Figure 8). These differences diminish as goals for net aquifer storage recovery increase, for example, from 1 MAF to 2 MAF in this case study.

### 3.4. Groundwater Salinity Changes

Evapotranspiration from crops consumes irrigation water, but not the salts in irrigation water. Here, deep percolation transports remaining irrigation water salts to the underlying undrained aquifer. At the end of the planning horizon, those salts are mixed with the available volume in the aquifer to compute changes in salinity for the next

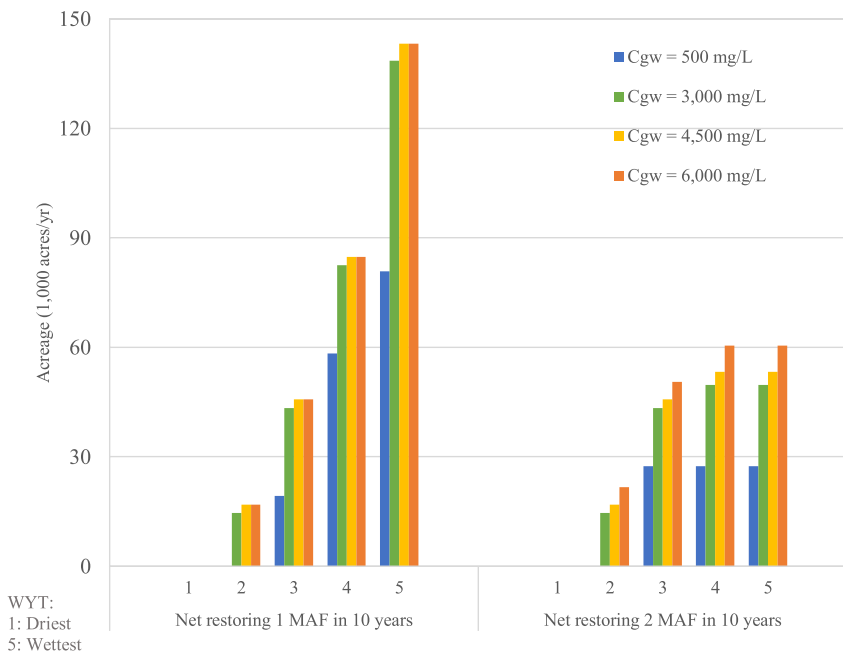



**Figure 8.** Annual crop acreage for each water year type (WYT) given different net recharge goals and different groundwater salinities.



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 2**
*The Change of 10-Year Groundwater Salinity (mg/L) Caused by Optimized Conjunctive Water Use and Cropping Decisions Often Increases Even Under No-Overdraft or Small Net Groundwater Storage Increase Goals, but Can Be Diluted by Substantial Net Recharge*

| Net recharge over planning horizon (MAF) | Initial aquifer salinity (mg/L) | | | | |
|---|---|---|---|---|---|
| | 500 | 1,500 | 3,000 | 4,500 | 6,000 |
| 0 | 239 | 239 | 239 | 199 | 125 |
| 1 | 172 | 81 | −55 | −191 | −328 |
| 2 | 116 | −50 | −300 | −550 | −800 |
| 3 | 69 | −162 | −508 | −854 | −1,200 |

planning horizon. Salt accumulation originates from concentrating salinity in pumped groundwater and from salts in applied surface water. Table 2 first shows two frustrating findings: if the aquifer does not naturally drain salts to streams or neighboring basins, (a) even with a management goal of "no overdraft," salts accumulate in groundwater; (b) even with a very demanding recharge goal (3 MAF/10 years in this case), groundwater salinity inevitably increases for fresher aquifers, which is common in the San Joaquin Valley and other drier regions with extensive irrigation (Pauloo et al., 2021).

Table 2 also shows that externally discharging irrigation water ("pump and dump" at salinities of 4,500 mg/L and 6,000 mg/L) with a balanced "no overdraft" management goal does not entirely avoid increasing groundwater salinity. But it reduces the rate of aquifer salination. Importantly, results show that under a net recharge management goal, an overall decrease in aquifer salinity can be achieved (if aquifer suffers from salinity), especially in more saline systems. These improvements neglect potential natural salt discharges (to neighboring basins and streams) from higher water levels. At lower groundwater salinity (≤1,500 mg/L), significant net aquifer recovery with fresher surface water is needed (≥2 MAF/10 years in our case study) to reverse groundwater salinity increases. The more impressive improvements in groundwater salinity with net groundwater storage recovery in high saline conditions result from reduced use of groundwater and larger contribution fresh surface water flows to groundwater recharge, when compared to intermediate salinity conditions.

### 3.5. Sensitivity Analyses

Here we examine the sensitivity of results to some parameters. The base case analysis has a "balanced" no-overdraft goal with initial groundwater storage of 10 MAF and 75,000 acres of incoming perennial crop. Lagrange multipliers show the effects of groundwater salinity changes on economic values of surface water and groundwater. The impacts caused by the changes in surface water availability also are explored.

#### 3.5.1. Lagrange Multipliers

In optimization modeling, Lagrange multipliers are a common by-product of solution algorithms and provide immediate sensitivity analysis results. In particular, Lagrange multipliers for each constraint in an optimization formulation indicate the change in the best value of the objective function for a small change in each constraint (Pierre, 1986). In this case, the Lagrange multipliers for conservation of mass of groundwater storage and surface water availability constraints indicate the overall economic value of having additional surface water and incremental changes in the groundwater storage change goal. In economics, Lagrange multipliers are often called "shadow prices" for each resource or constraint.

The value of surface water for each WYT is estimated by the Lagrange multipliers for the constraint in Equation 3, meaning the loss of net profit caused by one unit less surface water in each WYT. Sufficient groundwater storage also makes it possible to use a single Lagrange multiplier from conservation of groundwater mass constraint (Equation 4) to indicate the value of groundwater. These Lagrange multipliers provide insights on two concerns. First, for no-overdraft, we want to see how groundwater salinity alters the economic value of surface water and groundwater. Then, we compare the price of surface water and groundwater for different groundwater storage goals with and without groundwater salinity.

Table 3 (with zero net storage change) summarizes how initial groundwater salinity affects the economic value of surface water and groundwater for a no-overdraft goal. When groundwater salinity is low, groundwater has more economic value, as it can help support more perennial crops in dry years (when surface water is unavailable). With higher groundwater salinity, this drought buffering is unavailable and becomes a cost to agriculture, as irrigation return flows force additional groundwater pumping, which makes irrigation water too salty for perennial crops and reduces crop yield and profit. At the two highest groundwater salinities, groundwater is undesirable enough (in this example) that it is pumped for discharge outside the basin (having pumping and perhaps disposal costs without profit).

Salination of groundwater makes surface water more economically valuable in the driest years, as it reduces the aquifer's value for inter-annual, intermediate-term storage of surface water (via conjunctive use) and makes

19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 3**
*Economic Values of Water ($/AF) in Different Water Year Types (WYT) for Different Groundwater Recharge Goals With Different Initial Groundwater Salinity Under Base Climate*

| Initial groundwater salinity (mg/L) | Net recharge over planning horizon (MAF) | Surface water | | | | | Groundwater |
|---|---|---|---|---|---|---|---|
| | | WYT 1 | WYT 2 | WYT 3 | WYT 4 | WYT 5 | |
| 500 | 0 | 213 | 213 | 213 | 181 | 98 | 60.6 |
| | 1 | 282 | 282 | 224 | 184 | 161 | 97.4 |
| | 2 | 341 | 341 | 225 | 225 | 225 | 129 |
| 3,000 | 0 | 500 | 142 | 105 | 55 | 10 | −115 |
| | 1 | 855 | 216 | 189 | 150 | 95 | 64 |
| | 2 | 953 | 234 | 205 | 198 | 198 | 116 |
| 4,500 | 0 | 2,085 | 131 | 98 | 52 | 0 | −175 |
| | 1 | 2,536 | 212 | 186 | 147 | 91 | 62 |
| | 2 | 2,635 | 231 | 202 | 194 | 194 | 114 |
| 6,000 | 0 | 2,652 | 95 | 63 | 19 | 0 | −308 |
| | 1 | 3,387 | 212 | 186 | 147 | 91 | 62 |
| | 2 | 3,738 | 225 | 196 | 186 | 185 | 109 |

useable water scarcer overall. Surface water availability in the driest years limits the extent of profitable perennial crops, making farmers willing to pay more for surface water (coincidently at roughly the cost of desalted water in this case). The reason for such a high surface water value in the driest years is that when groundwater salinity can easily reduce perennial crop yield, less surface water means less irrigation water and/or higher irrigation water salinity when substituted with groundwater, which decreases perennial crop acreage and yield.

The range of economic values for surface water widens across wet and dry years as groundwater becomes more saline, driven by its disproportionate value to perennial crops. For low salinities, fresh groundwater availability dampens surface water price variation across years. But with more saline groundwater, variability in surface water's economic value increases, rising for dry years and declining, eventually to zero, for wetter years. Irrigating in wet years becomes a liability as its return flow to groundwater forces additional pumping of unwanted saline water. For wet WYTs, Equation 3 is no longer constrains as unused irrigation water is dumped.

Table 3 also summarizes Lagrange multiplier trends with and without salinity for different groundwater storage goals, supporting several findings: (a) although groundwater salinity reduces crop yields, groundwater becomes more valuable as aquifer recharge becomes harder with decreasing groundwater availability, and groundwater of higher salinity has lower value; (b) as aquifer recharge goals become more aggressive (meaning more water scarcity), the economic value of surface water in wetter years increases. The surface water in wetter years has higher value with lower groundwater salinity because deep percolation of surface water can be reused in drier years by groundwater pumping for more profitable perennial crops; and (c) the economic value of surface water in WYT 1 with groundwater salinity is much higher than in wetter years, and increases with groundwater salinity because less surface water in WYT 1 hurts perennial crop acreage and yields rather than less profitable annual crops in wetter years.

**Table 4**
*Water Year Type (WYT) Probabilities in Each Climate*

| WYT *j* | Climates | |
|---|---|---|
| | Drier | Base |
| 1 (Dry) | 0.25 | 0.2 |
| 2 | 0.25 | 0.2 |
| 3 | 0.2 | 0.2 |
| 4 | 0.2 | 0.2 |
| 5 (Wet) | 0.1 | 0.2 |
| Expected incoming surface water (TAF/yr) | 519 (−13%) | 599 |

### 3.5.2. Climate Effects

This sensitivity analysis changed the probabilities of WYTs to represent a drier climate. Table 4 summarizes the probability distributions of each WYT for each case.

A drier climate reduces agricultural profit. Comparing Tables 3 and 5 shows a drier climate increases the value of surface water in the driest year, especially with higher groundwater salinities, as drier years become more likely, and perennial crops are more vulnerable to drier years. However, because of

19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

**Table 5**
*Economic Values of Water ($/AF) From Base Climate to a Drier Climate in Different Water Year Types (WYTs) With Different Groundwater Salinities for No-Overdraft Goal*

| | Surface water | | | | | |
|---|---|---|---|---|---|---|
| Initial groundwater salinity (mg/L) | WYT 1 | WYT 2 | WYT 3 | WYT 4 | WYT 5 | Groundwater |
| 500 | 279 | 279 | 218 | 179 | 49 | 66 |
| 1,500 | 279 | 279 | 218 | 179 | 49 | 66 |
| 3,000 | 976 | 239 | 154 | 124 | 31 | 47 |
| 4,500 | 2,131 | 129 | 70 | 24 | 0 | −268 |
| 6,000 | 2,978 | 102 | 50 | 6 | 0 | −353 |

the higher probability of drier years and the much greater profitability of perennial crops, the value of surface water in wetter years declines (but remains positive).

When groundwater salinity is still "acceptable," overall useable dry-year water scarcity is less, and groundwater has more value, so when groundwater salinity is 3,000 mg/L, a drier climate turns groundwater value from negative (−$115/AF in Table 3) to positive ($47/AF in Table 5). However, if groundwater salinity is too high, groundwater reduces agricultural profitability for drier climates and groundwater value becomes more negative (e.g., for 4,500 mg/L, groundwater value drops from −$175/AF (Table 3) to −$268/AF (Table 5)).

However, a drier climate does not further change optimal conjunctive water use significantly in this example. The incoming perennial crop acreage in the base case is 75,000 acres, which is not so large that even with groundwater salinity of 6,000 mg/L, no incoming perennial crop is retired, as the resulted irrigation water salinity in drier years does not drive out existing perennial crops. When groundwater salinity is low, fewer new perennial crops are grown because less pumping occurs in drier years with higher probability (Yao, 2021). With high groundwater salinity, perennial crop acreage stays the same in a drier climate, with minor differences in second-stage decisions.

### 3.6. Trajectory of Groundwater Salinity and Groundwater Management

As discussed in Section 3.4, degradation of initial, good groundwater quality toward levels of salinity that reduce crop growth is inevitable as long as the aquifer lacks drainage of salinity to neighboring basins or streams, even without net drawdown of groundwater. More saline groundwater eventually shifts economically driven conjunctive water use from pumping in drier years to pumping in wetter years. More lower-value annual crops are planted in the wetter years with lower marginal profit, but fewer more profitable perennial crops can be maintained through drier years (Section 3.1).

If only short-term economic benefit is important, fertile lands will diminish in value as salinity rises in the soil and underlying aquifer without control. Perennial crops eventually become less profitable. This loss in profit does not even consider additional cost for treating saline "pump and dump" irrigation water prior to export from the basin. If we want to maximize expected net benefit in a longer term, aquifer recharge, though not the most profitable choice in any decade, is required to dilute the aquifer and increase water levels to create natural outflow of salts to streams or neighboring basins before groundwater salinity becomes too high for agricultural production. Our results show that economics eventually force a reversal of groundwater storage declines where salinity increases in the aquifer, even without regulatory measures.

Furthermore, given the exceedingly high economic value of surface water in dry years (Section 3.5), some treatment of saline groundwater might become profitable (typical cost is $1,000 to $2,000 per AF, depending on salt disposal costs). This also may affect the need for "dumping" groundwater in wet years. Additional modeling, accounting for treatment cost prior to irrigation, adding realistic costs for treating "dumped" irrigation water, and allowing more flexibility in (a) allocating crop-specific groundwater-surface water mixes of irrigation water and (b) including salt-tolerant crops among the annual crop category, may provide significantly different outcomes to the simplified cases explored here.

19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

## 4. Conclusions

In (semi)arid irrigated regions, with low groundwater salinity, more pumping occurs in drier years to support more perennial crops and smooth variability in annual crop acreage across years. However, as groundwater salinity rises enough to harm perennial crops, reduced pumping in drier years reduces higher-value perennial crop acreage, eliminates annual crops in drier years, and increases pumping in wetter years (to prevent excessive water levels) when annual crop acreage is already greater with much less marginal profit than perennial crops. In other words, annual crop acreage fluctuates more across years, with reduced expected profit. More saline groundwater can serve as a reservoir for drier years only to the extent it can be diluted by surface water, and agriculture becomes less profitable, smaller, and more variable across wetter and drier years.

Salinity accumulation has been a menace for irrigated agriculture from ancient Mesopotamia to the present day. Once groundwater overdraft is tamed, without the drainage of salts, accumulating groundwater salinity eventually profoundly shifts both groundwater management and overlying agriculture. With high salinity groundwater, groundwater must be "pumped and dumped" outside the basin to balance irrigation return flows to groundwater, which only imposes costs. So, fewer annual crops are grown in the wettest years compared to conditions with lower groundwater salinities to maintain perennial crop yield and avoid water logging. Additional research could consider options for treating groundwater prior to irrigation, to constrain "dumping" based on realistic discharge requirements (e.g., treatment cost), and to provide more flexibility in crop choice (annual crop portfolio with greater salinity tolerance) and in water allocation (crop-specific groundwater-surface water mix).

Though costly, earlier restoration of aquifer levels, with reduced pumping and increased surface water recharge, can slow groundwater salination and prolong the aquifer's value for agriculture. The importance of artificial recharge is further amplified when groundwater is already severely saline, when a mild aquifer recharge goal can be more profitable (if waterlogging is not an issue), by reducing the need to pump (and waste) groundwater and reducing limits on planting annual crops in wetter years.

## Appendix A: Detailed Derivations of Equation 1

This model completely mixes surface water (minus any artificial recharge) and groundwater for irrigating both perennial and annual crops. So the salt concentration of irrigation for WYT $j$ for decade $t$, $C_{\mathrm{irr},j,t}$ (since in this paper, only current decade is considered, so $t$ is omitted later), is a weighted average of surface water salinity, $C_{\mathrm{sw}}$, and initial groundwater salinity, $C_{\mathrm{gw},t-1}$.

$$C_{\mathrm{irr},j} = \frac{C_{\mathrm{sw}} \times (\mathrm{sw}_j - \mathrm{cap} \times X_{r,j,t}) + C_{\mathrm{gw},t-1} \times W_{p,j,t}}{\mathrm{sw}_j - \mathrm{cap} \times X_{r,j,t} + W_{p,j,t}} \tag{A1}$$

The electrical conductivity (EC) of irrigation water for WYT $j$, $\mathrm{EC}_{\mathrm{irr},j}$ can be derived from concentration $C_{\mathrm{irr},j}$ (Grattan, 2002):

$$\mathrm{EC}_{\mathrm{irr},j} = \begin{cases} \dfrac{C_{\mathrm{irr},j}}{640} & C_{\mathrm{irr},j} < 3,200 \\ 5 & 3,200 \geq C_{\mathrm{irr},j} < 4,000 \\ \dfrac{C_{\mathrm{irr},j}}{800} & C_{\mathrm{irr},j} \geq 4,000 \end{cases} \tag{A2}$$

**Table A1**
*Probability Transition Matrix of $Y_{p,j}$ Under Base Climate Scenario*

| $j_k \backslash j_{k+1}$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 0.5 | 0.5 | 0 | 0 | 0 |
| 2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| 3 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| 4 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |
| 5 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |

The salt stress coefficient of crop $i$ ($i = p$ for perennial crop, $a$ for annual crop) for WYT $j$, $K_{s-\text{salt},i,j}$, equals 1 if $\text{EC}_{\text{irr},j}$ is less than the EC threshold for crop $i$, $\text{EC}_{t,i}$, where yields decrease. Otherwise,

$$K_{s-\text{salt},i,j} = \begin{cases} 1, & \text{EC}_{\text{irr},j} < \text{EC}_{t,i} \\ 1 - \dfrac{b_i}{K_{y,i}} \left(\text{EC}_{\text{irr},j} - \text{EC}_{t,i}\right), & \text{EC}_{\text{irr},j} \geq \text{EC}_{t,i} \end{cases} \quad \text{for } i = p, a \tag{A3}$$

where $\frac{b_i}{K_{y,i}}$ is the decrease of crop $i$ yield per EC increment of the irrigation water, and $K_{y,i}$ is the crop $i$'s sensitivity to water stress (Allen, 2000). There is no $K_y$ value for almonds in FAO Irrigation and Drainage Paper No. 33 so we use a high value (1.2) to $K_{y,p}$, meaning the perennial crop is sensitive to water stress.

The combined salt and water stress coefficient of crop $i$ for WYT $j$, $K_{s,i,j}$, is the product of salt stress coefficient and water stress coefficient. Because stress irrigation is not considered here, $K_{s-\text{water}} = 1$.

$$K_{s,i,j} = \left(K_{s-\text{salt},i,j}\right) \times \left(K_{s-\text{water}}\right) = K_{s-\text{salt},i,j} \quad \text{for } i = p, a \tag{A4}$$

The relationship between yield decrease of crop $i$ and combined stress for WYT $j$ is:

$$1 - \frac{Y_{i,j}}{Y_{\text{max},i}} = K_{y,i}\left(1 - K_{s,i,j}\right) \quad \text{for } i = p, a \tag{A5}$$

Rearranging Equation A5, we can solve for actual yield of crop $i$ for WYT $j$, $Y_{i,j}$:

$$Y_{i,j} = \left(1 - K_{y,i}\left(1 - K_{s,i,j}\right)\right) Y_{\text{max},i} \quad \text{for } i = p, a \tag{A6}$$

For perennial and annual crop, $Y_{\text{max},i}$ in Equation A6 is $yld_p$ and $yld_a$, respectively. Similarly, $Y_{i,j}$ is represented by $Y_{p,j}$ and $Y_{a,j}$.

$$Y_{p,j} = \left(1 - K_{y,p}\left(1 - K_{s,p,j}\right)\right) yld_p \tag{A7}$$

$$Y_{a,j} = \left(1 - K_{y,a}\left(1 - K_{s,a,j}\right)\right) yld_a \tag{A8}$$

Here, perennial crop yield, $Y_{p,j}$, depends on the WYT because the blended irrigation water salinity varies across WYTs. We also consider the influence of drought (WYT 1) to represent a lagging effect of salinity in the root zone: If current year $k$ in current decade is WYT 1 (driest WYT), the yield for this year is $Y_{p,1}$. In year $k + 1$, no matter which WYT it is, perennial crop yield is reduced by the high salinity of irrigation in the previous year, such that there is a 50% of chance the yield is still $Y_{p,1}$ (the lowest yield), and 50% of chance the yield is $Y_{p,2}$ (the second lowest yield). On the other hand, if in current year $k$, we are not in WYT 1, the yield of perennial crop in year $k + 1$ is independent of the WYT in year $k$, thus being the same probability of the occurrence of each WYT

**Table A2**
*Probability of $Y_{p,j}$ in Year k, $P_{k,j}$ Under Base Climate Scenario*

| $k\backslash j$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 | 0.200 | 0.200 | 0.200 | 0.200 | 0.200 |
| 2 | 0.260 | 0.260 | 0.160 | 0.160 | 0.160 |
| 3 | 0.278 | 0.278 | 0.148 | 0.148 | 0.148 |
| 4 | 0.283 | 0.283 | 0.144 | 0.144 | 0.144 |
| 5 | 0.285 | 0.285 | 0.143 | 0.143 | 0.143 |
| 6 | 0.286 | 0.286 | 0.143 | 0.143 | 0.143 |
| 7 | 0.286 | 0.286 | 0.143 | 0.143 | 0.143 |
| 8 | 0.286 | 0.286 | 0.143 | 0.143 | 0.143 |
| 9 | 0.286 | 0.286 | 0.143 | 0.143 | 0.143 |
| 10 | 0.286 | 0.286 | 0.143 | 0.143 | 0.143 |



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$p_j$. So, perennial crop yield can be represented by a Markov chain. Take the normal climate scenario for example, ($p_j = 0.2, \forall j$), Table A1 is the transition matrix in normal climate scenario.

With a transition matrix in Table A1, we can derive the probability of $Y_{p,j}$ for year $k$, $P_{k,j}$, as Table A2. Ever since year 2, $Y_{p,j}$ has a higher probability to be $Y_{p,1}$ or $Y_{p,2}$ because of the toxic effect of salinity on the yield. Also, due to the independence between year $k$ and year $k+1$ and the same probability of each WYT under normal climate scenario, the probability of $Y_{p,3}$, $Y_{p,4}$, and $Y_{p,5}$ are the same for each year $k$.

Therefore, the expected net benefit of a perennial crop over 10 years, $B1$, is the sum of year $k$'s expected net benefits multiplied by discount factor ($k$ from 1 to 10), where the expected net benefit of year $k$ is the sum of the product of perennial crop profit (PMP quadratic cost function detailed in Yao (2021)) and the probability of $Y_{p,j}$ for year $k$, $P_{k,j}$.

$$\begin{aligned} B1 = & \frac{1}{(1+r)^{tT+1}} \sum_{j=1}^{5} P_{1,j} \left[ (v_p Y_{p,j}) X_{p,t} - \left( \alpha_p + \frac{1}{2} \gamma_p X_{p,t} \right) X_{p,t} \right] \\ & + \frac{1}{(1+r)^{tT+2}} \sum_{j=1}^{5} P_{2,j} \left[ (v_p Y_{p,j}) X_{p,t} - \left( \alpha_p + \frac{1}{2} \gamma_p X_{p,t} \right) X_{p,t} \right] + \ldots \\ & + \frac{1}{(1+r)^{tT+10}} \sum_{j=1}^{5} P_{10,j} \left[ (v_p Y_{p,j}) X_{p,t} - \left( \alpha_p + \frac{1}{2} \gamma_p X_{p,t} \right) X_{p,t} \right] \\ = & \sum_{k=1}^{10} \frac{1}{(1+r)^{tT+k}} \sum_{j=1}^{5} P_{k,j} \left[ (v_p Y_{p,j}) X_{p,t} - \left( \alpha_p + \frac{1}{2} \gamma_p X_{p,t} \right) X_{p,t} \right] \end{aligned} \tag{A9}$$

The expected net benefit of an annual crop over 10 years, $B2_t$, is the sum of yearly expected net benefits multiplied by discount factor, where the yearly expected net benefit is the product of annual crop profit in WYT $j$ and the probability of that WYT, $p_j$. It is a sum of a geometric series. Unlike the perennial crop, for any year $k$, the probability of annual crop with yield of $Y_{a,j}$ equals $p_j$.

$$\begin{aligned} B2 = & \frac{1}{(1+r)^{tT+1}} \sum_{j=1}^{5} p_j \left[ (v_a Y_{a,j}) X_{a,j} - \left( \alpha_a + \frac{1}{2} \gamma_a X_{a,j} \right) X_{a,j} \right] \\ & + \frac{1}{(1+r)^{tT+2}} \sum_{j=1}^{5} p_j \left[ (v_a Y_{a,j}) X_{a,j} - \left( \alpha_a + \frac{1}{2} \gamma_a X_{a,j} \right) X_{a,j} \right] + \ldots \\ & + \frac{1}{(1+r)^{tT+10}} \sum_{j=1}^{5} p_j \left[ (v_a Y_{a,j}) X_{a,j} - \left( \alpha_a + \frac{1}{2} \gamma_a X_{a,j} \right) X_{a,j} \right] \\ = & \frac{1-(1+r)^{-T}}{(1+r)^{tT} \times r} \sum_{j=1}^{5} p_j \left[ (v_a Y_{a,j}) X_{a,j} - \left( \alpha_a + \frac{1}{2} \gamma_a X_{a,j} \right) X_{a,j} \right] \end{aligned} \tag{A10}$$

## Appendix B: Calculation of Groundwater Salinity at the End of the Planning Horizon

The actual salt input from irrigation water in WYT $j$ can be calculated by multiplying actual amount of irrigation water in WYT $j$ (Equation B1) and salt concentration of irrigation water in WYT $j$, $C_{\text{irr},j}$ calculated from Equation A1. Equation B2 calculates the expected total amount of salt input from irrigation $\text{SALT}_{\text{irr}}$ during the entire planning horizon. Equations B3 and B4 calculates expected total salts from artificial recharge $\text{SALT}_{\text{AR}}$ and salts remaining in the aquifer $\text{SALT}_{\text{gw}}$, respectively. Finally, the expected concentration in groundwater at the end of planning horizon, $C_{\text{gw},t}$, is the sum of $\text{SALT}_{\text{irr}}$, $\text{SALT}_{\text{AR}}$, and $\text{SALT}_{\text{gw}}$, divided by the final groundwater storage (Equation B5).

$$\text{IRR}_{\text{Actural},j} = \text{aw}_p X_{p,t} + \text{aw}_a X_{a,j,t} \tag{B1}$$

$$\text{SALT}_{\text{irr}} = T \left( \sum_{j=1}^{5} p_j C_{\text{irr},j} \text{IRR}_{\text{Actural},j} \right) = T \left( \sum_{j=1}^{5} p_j C_{\text{irr},j} \left( \text{aw}_p X_{p,t} + \text{aw}_a X_{a,j,t} \right) \right) \tag{B2}$$



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$$\mathrm{SALT}_{\mathrm{AR}} = \mathrm{TC}_{\mathrm{sw}} \sum_{j=1}^{5} p_j \mathrm{cap} X_{r,j,t} \tag{B3}$$

$$\mathrm{SALT}_{\mathrm{gw}} = C_{\mathrm{gw},t-1}(\mathrm{GW}_{t-1} - T \sum_{j=1}^{5} p_j W_{p,j,t}) \tag{B4}$$

$$C_{\mathrm{gw},t} = \frac{\mathrm{SALT}_{\mathrm{irr}} + \mathrm{SALT}_{\mathrm{AR}} + \mathrm{SALT}_{\mathrm{gw}}}{\mathrm{GW}_t} \tag{B5}$$

## Appendix C: Case Study Parameters

| Symbol | Parameter (unit) | Value |
|---|---|---|
| $C_{\mathrm{sw}}$ | Salinity of surface water (mg/L) | 400 |
| $K_{y,p}$ | Perennial crop sensitivity to water stress | 1.2 |
| $K_{y,a}$ | Annual crop sensitivity to water stress | 1.1 |
| $\mathrm{EC}_{t,p}$ | Threshold EC with no perennial crop yield decrease (dS/m) | 1.5 |
| $\mathrm{EC}_{t,a}$ | Threshold EC with no annual crop yield decrease (dS/m) | 2.0 |
| $b_p$ | Perennial crop yield decrease per unit increment of EC (%) | 19 |
| $b_a$ | Annual crop yield decrease per unit increment of EC (%) | 7.3 |
| $T$ | Length of planning horizon (yr) | 10 |
| $L$ | Total available area (acre) | 500,000 |
| $H_o$ | Initial pump head (ft) | 200 |
| $B_o$ | Initial thickness of the aquifer (ft) | 200 |
| $s_y$ | Aquifer specific yield | 0.1 |
| $r$ | Constant discount rate | 0.035 |
| $\mathrm{ini}_p$ | Perennial crop initial establishment cost ($/acre) | 12,000 |
| $c_{\mathrm{land}}$ | Unit price of land for recharging ($/acre) | 300 |
| $c_{\mathrm{class1}}$ | Unit price of class 1 (firm contract) water ($/AF) | 42 |
| $c_{\mathrm{class2}}$ | Unit price of class 2 (surplus) water ($/AF) | 30 |
| class1 | Amount of firm contract water (TAF) | 500 |
| $c_e$ | Unit price of energy ($/kWh) | 0.189 |
| $\eta_p$ | Pumping efficiency | 0.7 |
| cap | Capacity of land for recharging (AF/acre/yr) | 15 |
| $1-\varphi$ | Irrigation efficiency | 0.85 |
| $t$ | Current planning horizon | 0 |
| $\mathrm{aw}_p$ | Annual applied water on perennial crop (ft/yr) | 4.07 |
| $\mathrm{aw}_a$ | Annual applied water on annual crop (ft/yr) | 4.84 |
| $yld_p$ | Annual maximum yield of perennial crop (ton/acre/yr) | 1 |
| $yld_a$ | Annual maximum yield of annual crop (ton/acre/yr) | 8 |
| $\upsilon_p$ | Unit price of perennial crop ($/ton) | 4226.68 |
| $\alpha_p$ | Perennial crop's intercept of PMP cost function ($/acre) | 1502.85 |
| $\gamma_p$ | Perennial crop's slope of PMP cost function ($/acre$^2$) | 0.00795 |
| $\upsilon_a$ | Unit price of annual crop ($/ton) | 157.28 |
| $\alpha_a$ | Annual crop's intercept of PMP cost function ($/acre) | 635.91 |
| $\gamma_a$ | Annual crop's slope of PMP cost function ($/acre$^2$) | 0.00294 |



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

## Data Availability Statement

Datasets for this research are preserved by the internet archive (https://web.archive.org/web/20170711215918/http:/water.ca.gov/economics/downloads/Models/SWAP_TechAppendix_080612_Draft.pdf). These datasets coupled with model formulation (v.1.0.0) and results are also available at: https://doi.org/10.5281/zenodo.5208677 and https://github.com/YiqingYao/Conjunctive_Salinity.

**Acknowledgments**
This research was partially supported by the US-China Clean Energy Research Center for Water-Energy Technologies (CERC-WET), Award #DE-1A0000018. This research has benefitted from discussions with Jonathan Herman and Samuel Sandoval Solis.

## References

Al Khamisi, S. A., Prathapar, S. A., & Ahmed, M. (2013). Conjunctive use of reclaimed water and groundwater in crop rotations. *Agricultural Water Management*, *116*, 228–234. https://doi.org/10.1016/j.agwat.2012.07.013

Allen, R. G. (2000). *Crop evapotranspiration: Guidelines for computing crop water requirements*. Food and Agriculture Organization of the United Nations.

Ayers, R. S., & Westcot, D. W. (1985). *Water quality for agriculture*. Food and Agriculture Organization of the UN.

Ben-Gal, A., Yermiyahu, U., & Cohen, S. (2009). Fertilization and blending alternatives for irrigation with desalinated water. *Journal of Environmental Quality*, *38*(2), 529–536. https://doi.org/10.2134/jeq2008.0199

CH2MHILL. (2012). DRAFT agricultural economics technical appendix. Retrieved from https://web.archive.org/web/20170711215918/http://water.ca.gov/economics/downloads/Models/SWAP_TechAppendix_080612_Draft.pdf

Coe, J. J. (1990). Conjunctive use—Advantages, constraints, and examples. *Journal of Irrigation and Drainage Engineering*, *116*(3), 427–443. https://doi.org/10.1061/(asce)0733-9437(1990)116:3(427)

CV-SALTS (2017). CV-SALTS releases new and innovative plan to address salt and nitrates in groundwater. Retrieved from https://www.cvsalinity.org/docs/committee-document/pubic-education-and-outreach-docs/3720-cv-salts-1-pager-fact-sheet-final/file.html

Fontanet, M., Fernàndez-Garcia, D., Rodrigo, G., Ferrer, F., & Villar, J. M. (2022). Combined simulation and optimization framework for irrigation scheduling in agriculture fields. *Irrigation Science*, *40*(1), 115–130. https://doi.org/10.1007/s00271-021-00746-y

Gorelick, S. M., Voss, C. I., Gill, P. E., Murray, W., Saunders, M. A., & Wright, M. H. (1984). Aquifer reclamation design: The use of contaminant transport simulation combined with nonlinear programing. *Water Resources Research*, *20*(4), 415–427. https://doi.org/10.1029/WR020i004p00415

Grattan, S. (2002). *Irrigation water salinity and crop production*. UCANR Publications.

Gupta, P., Khepar, S., & Kaushal, M. (1987). Conjunctive use approach for management of irrigated agriculture. *J. Agric. Eng*, *24*(3), 307–316.

Hazell, P. B. R., & Norton, R. D. (1986). Mathematical programming for economic analysis in agriculture.

Howitt, R., Kaplan, J., Larson, D., MacEwan, D., Medellín-Azuara, J., Horner, G., & Lee, N. (2009). *The economic impacts of Central Valley salinity*. University of California Davis, Final Report to the State Water Resources Control Board Contract.

Howitt, R. E. (1995). Positive mathematical programming. *American Journal of Agricultural Economics*, *77*(2), 329–342. https://doi.org/10.2307/1243543

Jalali, M. (2007). Salinization of groundwater in arid and semi-arid zones: An example from Tajarak, Western Iran. *Environmental Geology*, *52*(6), 1133–1149. https://doi.org/10.1007/s00254-006-0551-3

Kan, I., & Rapaport-Rom, M. (2012). Regional blending of fresh and saline irrigation water: Is it efficient? *Water Resources Research*, *48*(7), W07517. https://doi.org/10.1029/2011WR011285

Kaushal, M. P., Khepar, S. D., & Panda, S. N. (1985). Saline groundwater management and optimal cropping pattern. *Water International*, *10*(2), 86–91. https://doi.org/10.1080/02508068508686316

Khan, I. A. (1982). A model for managing irrigated agriculture1. *JAWRA Journal of the American Water Resources Association*, *18*(1), 81–87. https://doi.org/10.1111/j.1752-1688.1982.tb04531.x

Lefkoff, L. J., & Gorelick, S. M. (1990a). Benefits of an irrigation water rental market in a saline stream-aquifer system. *Water Resources Research*, *26*(7), 1371–1381. https://doi.org/10.1029/WR026i007p01371

Lefkoff, L. J., & Gorelick, S. M. (1990b). Simulating physical processes and economic behavior in saline, irrigated agriculture: Model development. *Water Resources Research*, *26*(7), 1359–1369. https://doi.org/10.1029/WR026i007p01359

Maas, E. V., & Grattan, S. (1999). Crop yields as affected by salinity. *Agricultural Drainage*, *38*, 55–108.

Malash, N. M., Flowers, T. J., & Ragab, R. (2008). Effect of irrigation methods, management and salinity of irrigation water on tomato yield, soil moisture and salinity distribution. *Irrigation Science*, *26*(4), 313–323. https://doi.org/10.1007/s00271-007-0095-7

Mandare, A. B., Ambast, S. K., Tyagi, N. K., & Singh, J. (2008). On-farm water management in saline groundwater area under scarce canal water supply condition in the Northwest India. *Agricultural Water Management*, *95*(5), 516–526. https://doi.org/10.1016/j.agwat.2007.12.010

Marques, G. F., Lund, J. R., & Howitt, R. E. (2010). Modeling conjunctive use operations and farm decisions with two-stage stochastic quadratic programming. *Journal of Water Resources Planning and Management*, *136*(3), 386–394. https://doi.org/10.1061/(ASCE)WR.1943-5452.0000045

Medellín–Azuara, J., Howitt, R. E., Lund, J. R., & Hanak, E. (2008). *The economic effects on agriculture of water export salinity south of the delta technical appendix I. Comparing futures for the sacramento-san joaquin delta*. Public Policy Institute of California.

Ortega-Reig, M., Palau-Salvador, G., Cascant i Sempere, M. J., Benitez-Buelga, J., Badiella, D., & Trawick, P. (2014). The integrated use of surface, ground and recycled waste water in adapting to drought in the traditional irrigation system of Valencia. *Agricultural Water Management*, *133*, 55–64. https://doi.org/10.1016/j.agwat.2013.11.004

Oster, J. D., & Grattan, S. R. (2002). Drainage water reuse. *Irrigation and Drainage Systems*, *16*(4), 297–310. https://doi.org/10.1023/A:1024859729505

Pauloo, R. A., Fogg, G. E., Guo, Z., & Harter, T. (2021). Anthropogenic basin closure and groundwater salinization (ABCSAL). *Journal of Hydrology*, *593*, 125787. https://doi.org/10.1016/j.jhydrol.2020.125787

Pierre, D. A. (1986). *Optimization theory with applications*. Courier Corporation.

Prendergast, J. B., Rose, C. W., & Hogarth, W. L. (1994). Sustainability of conjunctive water use for salinity control in irrigation areas: Theory and application to the Shepparton region, Australia. *Irrigation Science*, *14*(4), 177–187. https://doi.org/10.1007/BF00190189

Pulido-Velazquez, M., Marques, G. F., Harou, J. J., & Lund, J. R. (2016). *Hydroeconomic models as decision support tools for conjunctive management of surface and groundwater* (pp. 693–710). Integrated Groundwater Management.



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Rasouli, F., Kiani Pouya, A., & Šimůnek, J. (2013). Modeling the effects of saline water use in wheat-cultivated lands using the UNSATCHEM model. *Irrigation Science*, *31*(5), 1009–1024. https://doi.org/10.1007/s00271-012-0383-8

Rhoades, J. (1984). Use of saline water for irrigation. *California Agriculture*, *38*(10), 42–43.

Schoups, G., Addams, C. L., Minjares, J. L., & Gorelick, S. M. (2006). Sustainable conjunctive water management in irrigated agriculture: Model formulation and application to the Yaqui Valley, Mexico. *Water Resources Research*, *42*(10). https://doi.org/10.1029/2006WR004922

Schoups, G., Hopmans, J. W., Young, C. A., Vrugt, J. A., Wallender, W. W., Tanji, K. K., & Panday, S. (2005). Sustainability of irrigated agriculture in the San Joaquin Valley, California. *Proceedings of the National Academy of Sciences*, *102*(43), 15352–15356. https://doi.org/10.1073/pnas.0507723102

Sedki, A., & Ouazar, D. (2011). Simulation-optimization modeling for sustainable groundwater development: A moroccan coastal aquifer case study. *Water Resources Management*, *25*(11), 2855–2875. https://doi.org/10.1007/s11269-011-9843-9

Shang, S., & Mao, X. (2006). Application of a simulation based optimization model for winter wheat irrigation scheduling in North China. *Agricultural Water Management*, *85*(3), 314–322. https://doi.org/10.1016/j.agwat.2006.05.015

Singh, A. (2012). An overview of the optimization modelling applications. *Journal of Hydrology*, *466*, 167–182. https://doi.org/10.1016/j.jhydrol.2012.08.004

Singh, A. (2014). Conjunctive use of water resources for sustainable irrigated agriculture. *Journal of Hydrology*, *519*, 1688–1697. https://doi.org/10.1016/j.jhydrol.2014.09.049

Tyagi, N., & Narayana, V. D. (1981). Conjunctive use of canals and aquifers in alkali soils of Karnal. *Journal of Agricultural Engineering of New Delhi, India*, *18*, 78–91.

Tyagi, N. K. (1986). Optimal water management strategies for salinity control. *Journal of Irrigation and Drainage Engineering*, *112*(2), 81–97. https://doi.org/10.1061/(asce)0733-9437(1986)112:2(81)

Tyagi, N. K. (1988). Managing salinity through conjunctive use of water resources. *Ecological Modelling*, *40*(1), 11–24. https://doi.org/10.1016/0304-3800(88)90099-3

Tyagi, N. K., & Narayana, V. V. D. (1984). Water use planning for alkali soils under reclamation. *Journal of Irrigation and Drainage Engineering*, *110*(2), 192–207. https://doi.org/10.1061/(asce)0733-9437(1984)110:2(192)

Yao, Y. (2021). *Managing groundwater for agriculture, with hydrologic uncertainty and salinity*. Ph.D., University of California. Retrieved from https://www.proquest.com/docview/2533283591/abstract/DDC58BC1AFE442DBPQ/1

Yeh, W. W.-G. (1985). Reservoir management and operations models: A state-of-the-art review. *Water Resources Research*, *21*(12), 1797–1818. https://doi.org/10.1029/WR021i012p01797

Zhu, T., Marques, G. F., & Lund, J. R. (2015). Hydroeconomic optimization of integrated water management and transfers under stochastic surface water supply. *Water Resources Research*, *51*(5), 3568–3587. https://doi.org/10.1002/2014WR016519



19447973, 2022, 10, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021WR031058 by Amrita Vishwa Vidyapeetham, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License